

Principal component analysis of dietary and lifestyle patterns in relation to risk of subtypes of esophageal and gastric cancer

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Purpose To perform pattern analyses of dietary and lifestyle factors in relation to risk of esophageal and gastric cancers. **Methods** We evaluated risk factors for esophageal adenocarcinoma (EA), esophageal squamous cell carcinoma (ESCC), gastric cardia adenocarcinoma (GCA), and other gastric cancers (OGA) using data from a population-based case-control study conducted in Connecticut, New Jersey, and western Washington state. Dietary/lifestyle patterns were created using principal component analysis (PCA). Impact of the resultant scores on cancer risk was estimated through logistic regression. **Results** PCA identified six patterns: meat/nitrite, fruit/vegetable, smoking/alcohol, legume/meat alternate, GERD/BMI, and fish/vitamin C. Risk of each cancer under study increased with rising meat/nitrite score. Risk of EA increased with increasing GERD/BMI score, and risk of ESCC rose with increasing smoking/alcohol score and decreasing GERD/BMI score. Fruit/vegetable scores were inversely associated with EA, ESCC, and GCA. **Conclusions** PCA may provide a useful approach for summarizing extensive dietary/lifestyle data into fewer interpretable combinations that discriminate between cancer cases and controls. The analyses suggest that meat/nitrite intake is associated with elevated risk of each cancer under study, while fruit/vegetable intake reduces risk of EA, ESCC, and GCA. GERD/obesity were confirmed as risk factors for EA and smoking/alcohol as risk factors for ESCC.